

TANVIR HASAN

Backend Software Engineer | Cloud-Native & Distributed Systems

Dhaka, Bangladesh · tanvir.hasan.rafi.00@gmail.com · +8801869494543 · [LinkedIn](#) · [GitHub](#) · [Portfolio](#)

SUMMARY

Backend engineer with 3+ years building event-driven, cloud-native systems on Azure for US-based enterprise clients, owning work from technical design through production support. Also builds multi-agent AI systems with orchestration, tool calling, and LLM-based workflows.

PROFESSIONAL EXPERIENCE

Brain Station 23

Dhaka, Bangladesh | March 2023 - Present

Software Engineer II (promoted from Software Engineer Intern)

- Design and build **event-driven backend systems** for a US-based industrial IoT client using Azure Functions and IoT Event Hub, ingesting real-time telemetry from devices across North America at **millions of events per day**.
- Improved **Azure SQL query performance by 55%** through execution-plan analysis, indexing strategy, query refactoring, caching, and connection tuning – cutting response latency for enterprise workloads.
- Built fault-tolerant distributed messaging workflows on **Azure Service Bus** with retry policies, dead-letter handling, and failed-message isolation, improving reliability of mission-critical processing.
- Instrumented services with **Azure Monitor and Application Insights**, building operational dashboards that surface performance bottlenecks and shorten time-to-diagnosis on production incidents.
- Debug and resolve complex production issues via root-cause and performance analysis, shipping fixes that reduce recurring failures and stabilize long-running services.
- Review pull requests and mentor junior engineers on architecture, testing, and maintainability, while translating business requirements into scalable designs with stakeholders across the USA and Bangladesh.

PROJECTS

AI Multi-Agent Orchestration Platform

Python | FastAPI | LLMs | MCP | Tool Calling

- Designed and implemented a custom **multi-agent AI platform** with a central orchestration layer that routes tasks to specialized agents and dynamically invokes tools based on task requirements.
- Built agent workflows that **select and invoke appropriate tools**, process results, and pass structured outputs back through the orchestration layer for downstream agents.
- Exposed the platform over a **FastAPI** service with MCP-based tool integration, making agent capabilities reusable across multiple client applications.

TECHNICAL SKILLS

Languages: TypeScript (Node.js), Python, C#, Go

Backend: RESTful APIs, Microservices, Event-Driven Architecture, Message Queues, Background Services, API Versioning **Cloud &**

DevOps: Azure (Functions, Service Bus, IoT Event Hub, Monitor, API Management, Key Vault), Docker, Kubernetes, CI/CD

Databases: Azure SQL Server, PostgreSQL, MongoDB, Redis – query optimization, indexing, schema design **System**

Design: Distributed Systems, Scalability, Fault Tolerance, Observability, Clean Architecture, SOLID **Practices:**

Agile/Scrum, Test-Driven Development, Code Review, Performance Optimization, Technical Mentorship

EDUCATION

B.Sc. in Computer Science & Engineering

2022

University of Information Technology and Sciences (UITS) | Dhaka, Bangladesh

LEADERSHIP & ACHIEVEMENTS

- University Computer Club** – Led a 200+ student community and a team of 10+ executives, mentoring students in competitive programming.
- Inter-University Programming Contest** – Directed a 350+ participant contest across 10 universities, managing sponsorships and coordination.